NTNU

Department of Industrial Economics and Technology Management

Examination paper for TIØ4145 Corporate Finance

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Informasjon om trykking av eksamensoppgave

Originalen er:

1-sidig ☐ **2-sidig** ☐

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Task 1 – CAPM [15%]

Assume the CAPM model is valid. You are given the following information:

Stock	Risk free rate	Expected return on the market portfolio	Expected return on the stock	β	Price of the stock in period 0	Expected price of the stock in period 1
A	10%	20%	10%		100	
B	10%	20%		1.5		125
C	10%		20%	2.0	200	

The stocks pay no dividends next period. This is enough information to complete the table. Calculate all the missing data.

Task 2 – International Corporate Finance [20%]

Consider a Norwegian investor who is planning to invest in a South African risk-free security that pays out 480 RAND in one year. Assume that the risk-free interest rate in South Africa is 4.6%. The current spot exchange rate of Norwegian Kroner to South African Rand is 1.66 RAND/NOK

a) What will the investor pay for this security in NOK?

Assume the risk-free interest rate in Norway is 3% and the one-year forward rate is 1.6 RAND/NOK.

b) Is it possible for the Norwegian investor to profit from an arbitrage opportunity? If yes, illustrate this opportunity and calculate the arbitrage profit. If no, show that the arbitrage profit is zero. Start the illustration by assuming that an investor borrows 100 NOK.
c) Under the assumptions in this task, we can conclude that these markets are not internationally integrated. Why is that?
d) Find the forward rate that would be implied by the assumption of internationally integrated capital markets.

Task 3 – Capital Structure [20%]

ABC Inc. is an all-equity firm with 10 million shares outstanding which trade at 15 NOK per share, and if faces a 40% corporate tax rate. Although investors currently expect ABC to remain an all-equity firm, ABC plans to announce tomorrow that it will issue perpetual debt for an amount of NOK 80 million and will use all the funds to repurchase shares. Market participants understand that with this new level of debt ABC could become financially distressed some time in the future. They also understand that the present value of the expected financial distress costs borne by shareholders is NOK 40 million. The announcement will also convey that ABC has no further plans to change its debt level.

a) What is the market value of ABC's assets before the announcement?
b) What is the market value of ABC's assets after debt is issued and the shares are repurchased? What is the market value of ABC's equity?
c) What is the market value of ABC's equity and the share price upon announcement (but before execution) of the transaction?
d) How many shares will ABC repurchase?
e) Is this change in capital structure beneficial to shareholders? Why/why not?

Task 4– Option Valuation [30%]

Consider a stock with a current price of 400 NOK. In each of the next two years (2019 and 2020) the stock price will either increase by 25% or decrease by 20%. The stock pays no dividends and the risk-free interest rate is equal to 7%. Use a two-period binomial model to answer the following questions

a) Calculate the price of a European call option on this stock with a strike price of 375 NOK.

Assume now that the stock pays a 25% dividend in period 2019.

b) Recalculate the price of the European option in a) given the new assumption regarding dividends. (***hint:*** you need to calculate the option values from 2019 onwards based on ex dividend price, whereas the option value in 2018 based on the cum dividend price)
c) Compare the prices in a) and b) and explain the difference (or the absence of difference).
d) What is the price of the American option on the same stock as in b) with the same strike price of 375 NOK? (the same ***hint*** as in b) applies).
e) Compare the prices of the American and the European options in b) and d) and explain the difference (or the absence of difference).

Note: you can choose between replicating portfolio method or risk-neutral probabilities to price the options.

Task 5 – Law of One Price [15%]

In the discussions of some of the models in this course, we relied on the no-arbitrage argument or the Law of One Price: "If two investment strategies have the same payoff tomorrow, they must have the same price today."

For the following theories/relationships, **a)** shortly explain each theory/relationship and **b)** explain which of the following theories relies explicitly on no-arbitrage arguments.

i. The relationship between the value of coupon bonds and zero-coupon bonds. Use an example to illustrate the answer to this question (you need to make some assumptions about coupon payments, time to maturity and face value).
ii. The Valuation of Stocks based on Multiples.
iii. The Fama-French-Carhart multifactor model of risk.